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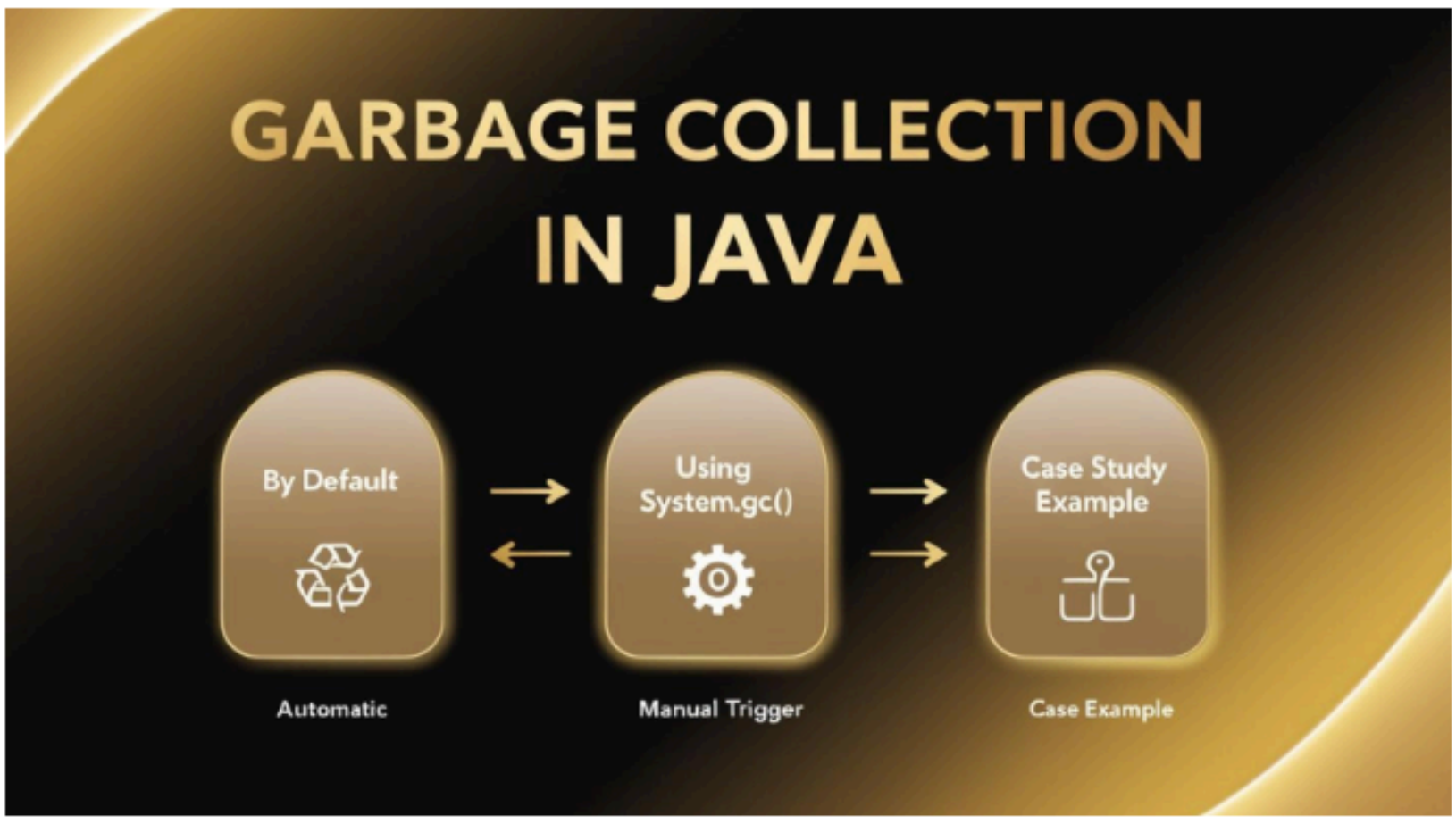
WHAT IS GARBAGE COLLECTION IN JAVA? 🗑️♻️



- Memory Management Technique in Java.
- Technically, it is a memory deallocation process in Java.
- In Java, the JVM (Java Virtual Machine) is responsible for memory management.
- JVM allocates and deallocates memory based on our Java code.
- Java has a Garbage Collector that is part of the JVM.
- When the JVM runs, the garbage collector identifies unused objects and removes them from memory.
- It removes unused objects from the heap memory, which is called garbage collection.
- This entire process is referred to as garbage collection.

Example:

- Let's say you have a phone with 128 GB memory and lots of unwanted videos.
- You delete the unwanted videos to manage the phone's memory.
- Similarly, garbage collection is a process of cleaning up unwanted memory in a Java application.

NOTE: 📝

- Using the new operator, we can allocate memory for a new object.
- We can manually trigger garbage collection using System.gc().

CASE STUDY BASED QUESTIONS

Case Study Based Questions: Different Ways to Achieve Garbage Collection ♻️

#1: By Default 🔄

- ✅ Garbage collection happens automatically.
- 🛠️ The garbage collector is a part of the JVM.
- 🧹 By default, the garbage collector periodically checks the heap memory to:
  - Identify unused objects.
  - Remove them from memory.

#2: Using System.gc() ⚙️

- 🖥️ You can request the JVM to call the garbage collector manually.
- 💻 Use the System.gc() method to perform garbage collection when needed.

#3: Have You Ever Used System.gc()? 🤔

- "I never used it directly."
- Instead, I rely on: finally block.
- try-with-resources for better resource management.

Case Example: (Based On Real time Experience)

- Once, I faced a memory issue where I had to increase the heap size.
- In server configuration, I updated the -Xms<size> property.
- This helped me fix an OutOfMemoryError issue.

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BASIC RULE FOR GARBAGE COLLECTION

Basic Rule for Garbage Collection 📋

How does the Garbage Collector pick objects?

- It picks, Nullified objects.
- Dereferenced objects.
- Objects that are no longer needed after some operation.

Example:

- If an object is no longer required, we can nullify it so it can be picked up by the garbage collector during the process.

Best Practice: ✅

- Yes, it is a good practice to nullify objects that are no longer needed in your Java code.
- This helps optimize memory management and improves the performance of your application.

Have You Ever Faced Out Of Memory in Your Application? 🛠️💻

- ✅ Yes, I have faced it.
- 🧠 The memory allocated for the heap was insufficient.

🔧 **Solution:**

- I updated the heap configuration to resolve the issue.
- Increased the maximum heap size in the server configuration.

📋 **Details:**

Used the JVM properties:

- -Xms<initial size> (initial heap size).
- -Xmx<max size> (maximum heap size).
- Updated the -Xmx property to allocate a larger max size for the heap.

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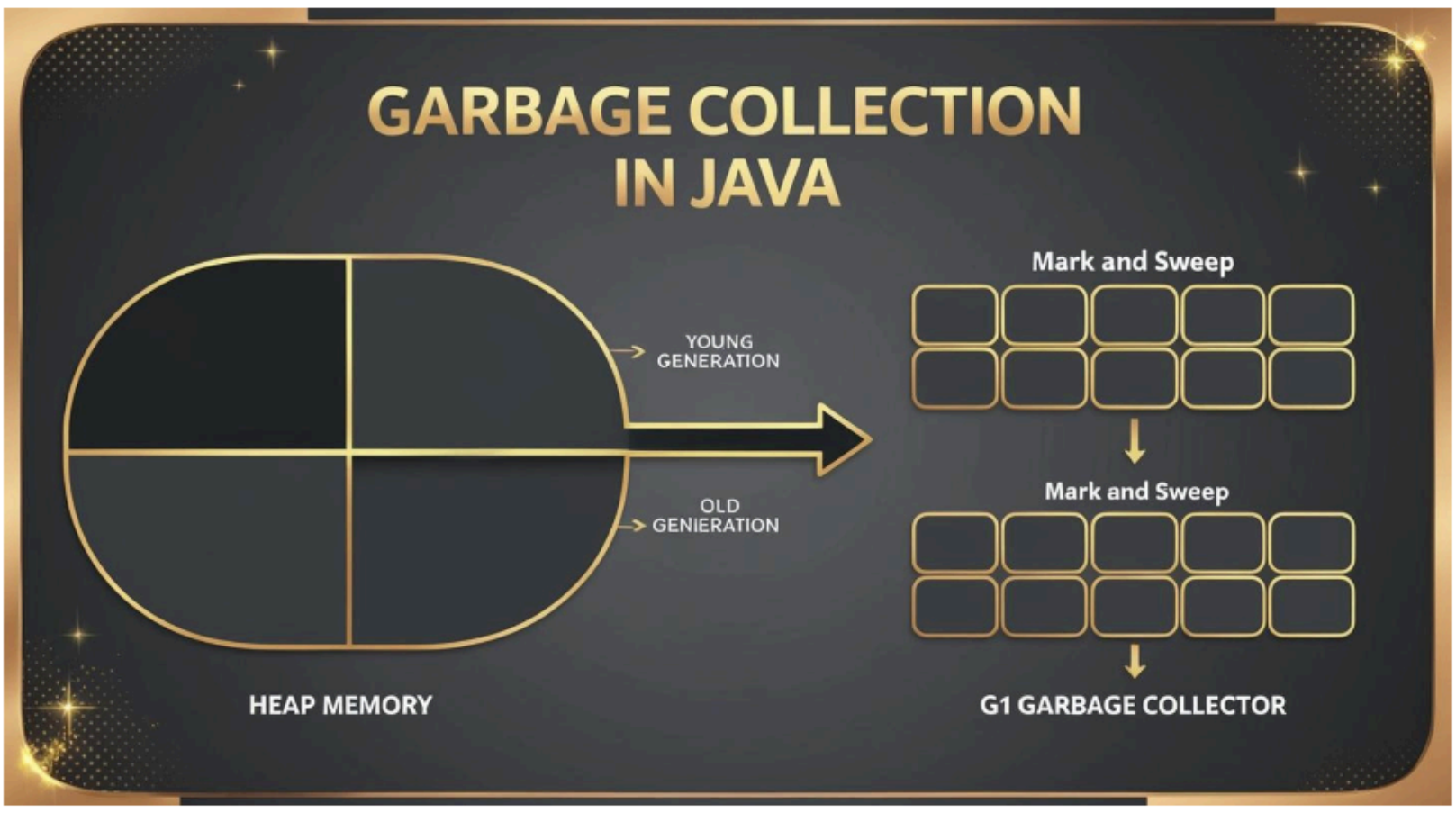
GARBAGE COLLECTION ALGORITHMS IN JAVA 🗑️♻️

Garbage Collection algorithms in Java 🗑️♻️



Mark and Sweep Algorithm 🗑️🧹

- ✅ Mark Phase: JVM marks all active objects in the heap.
- 🚮 Sweep Phase: JVM frees up memory occupied by unmarked (inactive) objects.
- ⚠️ Limitation: Not suitable for huge heaps.

G1 Garbage Collector 🚀🛠️

- Designed specifically for large heaps.
- 🚀 Improves execution time for large memory spaces.
- 🔄 Divides the heap into smaller regions to optimize memory management.
- ⚙️ Uses concurrent technology to deallocate memory in parallel.

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PURPOSE OF THIS DOCUMENT

- Use this PDF as a quick last-minute revision guide only.

- For more details, please perform R&D and gain a deeper understanding of how memory architecture works.

- Once the interview series is complete, we will cover this topic in detail.